

Week 1 Logbook

Objective

Learn LTspice, simulate switch circuits and diode circuits, understand waveform behaviour, and study basic semiconductor devices through datasheets and simulation.

SW-A: Series Switch

Measured:

- Voltage across switch
- Output voltage
- Current through resistor

Compared ON and OFF states.

Learning

A closed switch behaves like a short circuit and therefore drops almost no voltage. An open switch blocks current and therefore almost the entire supply voltage appears across it.

SW-B: Shunt Switch

Challenge

The measured switch current appeared negative in LTspice.

Initially assumed current was physically flowing in the reverse direction.

Investigation

Compared resistor current and switch current.

Examined node voltages and current paths.

Learning

The negative sign was due to LTspice's current reference direction. Current direction did not reverse; only the measurement reference changed.

Although SW-A and SW-B both showed approximately 5 mA current in the ON state, the output behaviour was completely different:

- SW-A ON → Output HIGH
 - SW-B ON → Output LOW
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SW-C: Series + Shunt Switch

Investigation

Analysed four switch combinations:

- ON/OFF
- OFF/ON
- OFF/OFF
- ON/ON

Tracked voltage at multiple nodes.

Learning

Observed:

- HIGH state
 - LOW state
 - Floating state
 - Shoot-through condition
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Half-Wave Rectifier

Investigation

Compared waveforms before and after the diode.

Learning

The diode conducts during positive half cycles and blocks during negative half cycles.

The negative half-cycle is removed from the output waveform.

Full-Wave Bridge Rectifier

Challenge

Found it difficult to understand why the measured source waveform did not appear sinusoidal without the filter.

Full-Wave Rectifier with Filter Capacitor

Expected to see a full-wave rectified waveform after adding the capacitor.

Instead observed an almost constant DC voltage.

Investigation

Compared output with and without capacitor.

Learning

The capacitor stores charge and discharges slowly between peaks, smoothing the pulsating DC into a nearly constant DC output.

This clarified the practical purpose of filter capacitors.

Zener Shunt Regulator

Challenge

Initially viewed the Zener diode as just another diode.

Did not understand why it was connected in reverse bias.

Investigation

Studied Zener operation and datasheet parameters.

Learning

Unlike a normal diode, a Zener diode is intentionally operated in reverse breakdown.

It maintains an approximately constant output voltage and can therefore be used for voltage regulation.

Diode Clipper

Learning

Changing diode orientation changes which half cycle is clipped.

The clipper limits voltage by removing part of the waveform.

Diode Clamper

Learning

A clamper does not remove part of a waveform.

Instead, it shifts the entire waveform by adding a DC offset while preserving waveform shape.

This clarified the distinction between clipping and clamping.

LED

Learning

Instead of choosing resistor values arbitrarily, resistor selection should be based on:

- Forward voltage
- Desired operating current
- Maximum current rating

This was applied to calculate the LED current-limiting resistor.

